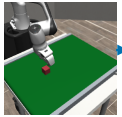
D reference: auxiliary supervision, action-only inference



agentview RGB, 224²

P15 dev84 / state 010

DINOv3
frozen

DINO → 768 + view embedding
+ Linear(depth: 1 → 768)
+ Linear(XYZ: 3 → 768)
sum per visual patch

depth: mean metric patch depth (m)

XYZ: fixed patch centers + depth + K + camera pose → robot base

single view
196 visual

S: scene bank
196 current visual tokens
+ 8 EEF-history tokens
+ 8 GT kinematic tokens

Token banks

- scene
- meta — **16 learned meta**
- flow — noised F + u (train)
- action — noised actions + u

3DRoPE: current visual Q/K
+ matching flow-query XYZ only

Shared attention + split FFNs × 4

shared QKV / O
rows query; cols key

	S	M	F	A
S	■			
M	■	■		
F	■		■	
A		■		■

access mask

Online: S + M + A; omit F
A reads M + A; no F → A edge

world/meta FFN

flow FFN

action FFN

Readouts and execution

Auxiliary objectives (training)

pool 16 meta deterministic pose → **Lpose** ← GT pose + masks 50–400 ms

scene-flow FM field → **LFM** ← GT flow + masks

action FM field → **Euler ×5**

predict 8 → execute 1 → replan

Solid: data; dashed: supervision